Master's Portfolio Report

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Abstract

This report conducts a longitudinal analysis on the results of the boys and girls cross country teams at Leland High School in order to determine if there is significant variation between years, and if other factors like weather have an effect. This can aid in possibly predicting future performance of the team. The results show no significant difference in both teams across years. However, the athlete's grade level and their ability to improve do seem to have an effect on their results. Ultimately, any patterns in the results are most likely better explained by other variables not used in this analysis, such as an athlete's physical statistics.

1 Introduction

Sports teams are known for having fluctuations in performance over the years, which can be even be grouped into particular "eras".[9] While part of it can be attributed to random variance, some of it is likely related to observable factors. The understanding that develops between members of the same team over time [1][6], pivotal leadership such as Coach K's legendary reign over the Duke basketball team [7], and perhaps even socioeconomic conditions can all affect the team's performance. Because of this, it can be helpful from a coach's perspective to be able to understand and possibly even predict these variations in performance. While teams that compete in football or basketball on the national level may be more well known, this project's focus is the cross country team at Leland High School in San Jose, CA due to prior experience and knowledge.

The Leland cross country team is a co-ed sports team that is active from August to November each year. The team participates in competitions ("meets") with other schools on the weekdays and more casual "invitationals" on the weekends in September and October, as well as a championship meet to determine who advances to higher-level competition in early November. Since 2002, the team has consistently recorded yearly results and held all their meets at the course Montgomery Hill (2.74 Miles), located in San Jose, as well as having the championship meet and occasional invitational at Crystal Springs (2.95 Miles), located in Belmont, CA. The one exception to this is the year 2020 having no results on these courses due to the COVID-19 pandemic. During this time, the team has also had the same head coach. Due to this consistency, I decided to focus on the years starting from 2002 up to the most recent season 2025.

The girls' and boys' teams race separately, and both are further divided into varsity, junior varsity (JV), and reserve teams. The varsity and JV teams are the ones that compete for points and are also restricted to seven people per team, although the exact lineup often varies across meets. Points are scored based on the place that the first five members of the team get; to win, as many team members should finish the race as fast as possible.[3] Since athletes on opposing teams do not generally interfere with each other directly, winning does not require intensive strategy beyond being fast, so cross country might not be considered a "team sport" in the traditional sense.[5] However, members of a team can somewhat influence each other by keeping pace with each other, and having a fast group of individuals can set higher standards for the rest of the team. Because of this, it can be more effective to look at the team's performance as a whole rather than splitting it into the different varsity levels, and there is cause to suspect that team performance may wax and wane each year as old cohorts leave and new ones emerge.

2 Methods

2.1 Data Collection

The data for this project was scraped from athletic.net [8], a publicly available website where schools will post their cross country results. A currently existing Python scraper on GitHub made specifically for this website was used and slightly modified for the data collection [4].

The weather data for each course consisted of the temperature in degrees Fahrenheit and relative humidity. This was collected in Python using the meteostat module, which gets data from weather stations

closest to a particular geographical point specified by longitude and latitude. For each race date, the data from the closest weather station with both temperature and humidity available. The exception to this rule is for Crystal Springs, as San Francisco and Half Moon Bay are weather stations that are close by, but as they are known for being significantly cooler than other regions of the Bay Area [2], data from them was not used.

2.2 EDA

In the data, a **result** is the response variable of time in minutes, and is indexed at the (athlete, date, distance) level. The smaller a result is, the faster the athlete completed the race. Besides the response, the data includes the predictors of athlete name, the date of a result, the grade of the athlete when they achieved a result, the distance of a race in either meters or miles, and whether each result of a specific athlete was their best time ever achieved (PR), best time in that season (SR), or improvement over the last race at the same distance. The observational units for this project are at the athlete level. Additionally, only athletes who had results at either the 2.74 or 2.95 mile distances were included in the dataset. To consolidate the effects of athlete improvement over time, a **"percent improvement"** variable was defined as $100 * [(cumulative_mean(time)) - current_time)/cumulative_mean(time)]$ per distance for each athlete and set to 0 for the athlete's first race at the distance. The figures below show the relationships between the main predictors and the response. In particular, percent improvement seems to be around 10% for the fastest athletes with increasing magnitude in both directions as times get slower. Intuitively, this may represent how it is harder to get faster on a particular course once you reach top speed, and athletes below this peak speed have more room to get both faster and slower.

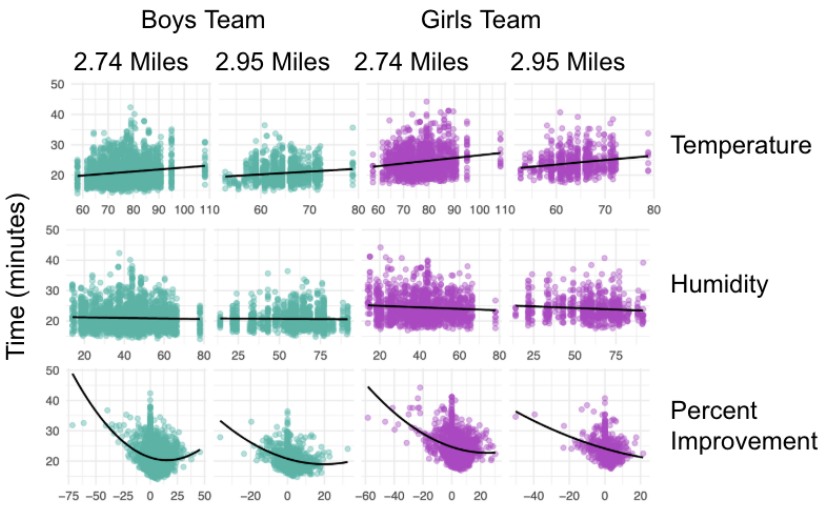

Figure 1: Quantitative predictors temperature, humidity, and percent improvement compared to time.

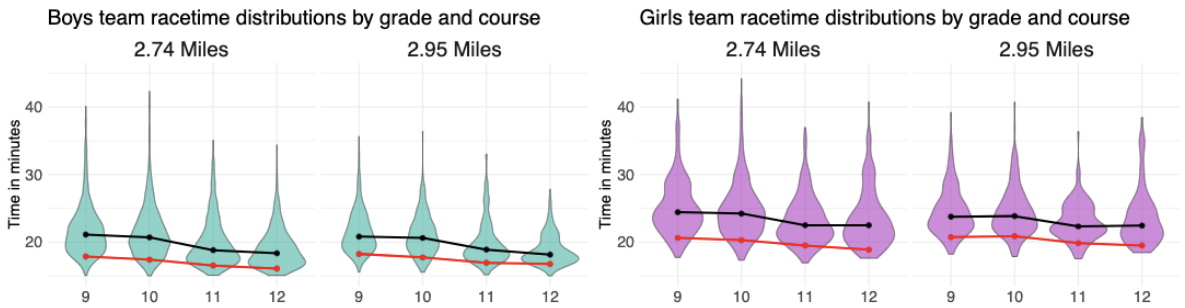

Figure 2: Times tend to improve as athletes get older. Lines at top 50th (black) and 10th (red) percentiles.

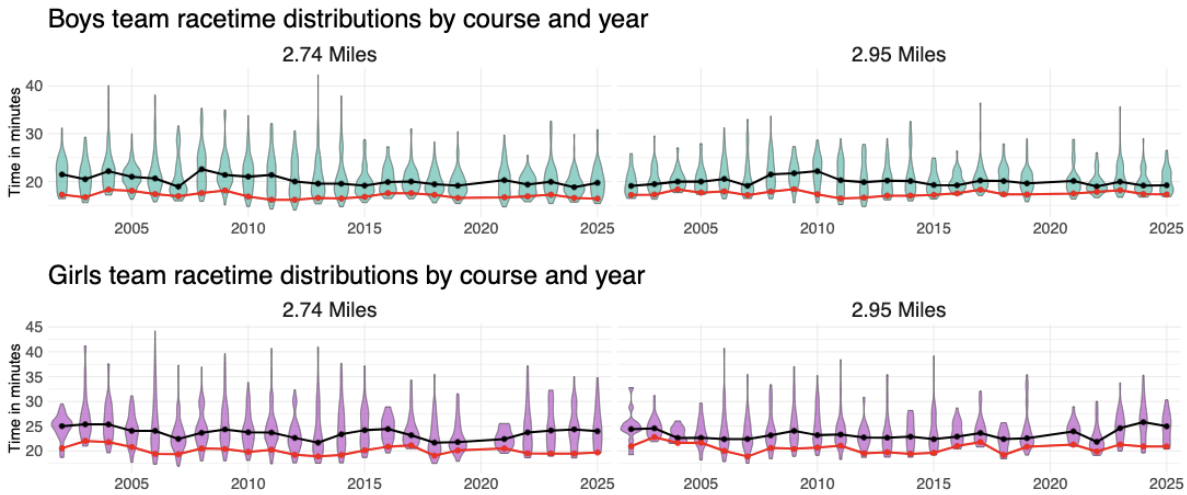

Figure 3: Slight variation in times across seasons, most notably for the girls' team at 2.74 Miles.

2.3 Model

As the data consisted of repeated observations from the same athletes and was organized in a athlete → distance → result structure, a linear mixed effects longitudinal hierarchical model was used:

$$y_{ijk} = \beta_0 + \beta_1 grade_{ijk} + \beta_2 time_pct_imp_{ijk} + \beta_3 temp_{jk} + \beta_4 rhum_{jk} + \beta_5 race_year + \beta_6 distance_j + b_i + \epsilon_{ijk}$$

The outcome of time in minutes, y_{ijk}, is indexed by the athlete i at distance j and date of the race k. The $time_pct_imp_{ijk}$ variable is defined as the percentage improvement of result ijk over the cumulative mean of athlete i's previous results at distance j. The model uses fixed effects for grade, percent improvement ($time_pct_imp$), temperature ($temp$), relative humidity ($rhum$), the year of the race ($race_year$), and the race distance ($distance$), and a random intercept for athlete (b_i).
Using the lme4 package in R, the girls and boys teams were modeled separately with the same model as their times are obviously going to be different and the difference between them is not of importance. Additionally, the other variables might have interactions with gender, making the model more complicated.

2.3.1 Model Assumptions

The assumptions of this model are that the relationship between the predictors and the response is linear, the random intercept $b_i \sim N(0, \tau^2)$ and is independent of both the random errors ϵ_{ijk} and fixed effects, and that the random errors are independent, are centered around zero with constant variance and are normally distributed for inference. In order to assess the model assumptions and performance, the plots of residuals versus fitted values and normal Q-Q plots for the residuals will be checked, as well as the significance of the coefficients.

3 Results

Plots of the residuals of the model versus fitted values, the normal Q-Q plot of the residuals, and quantification of the uncertainty of the estimated fixed and random effects for the longitudinal model are shown below.

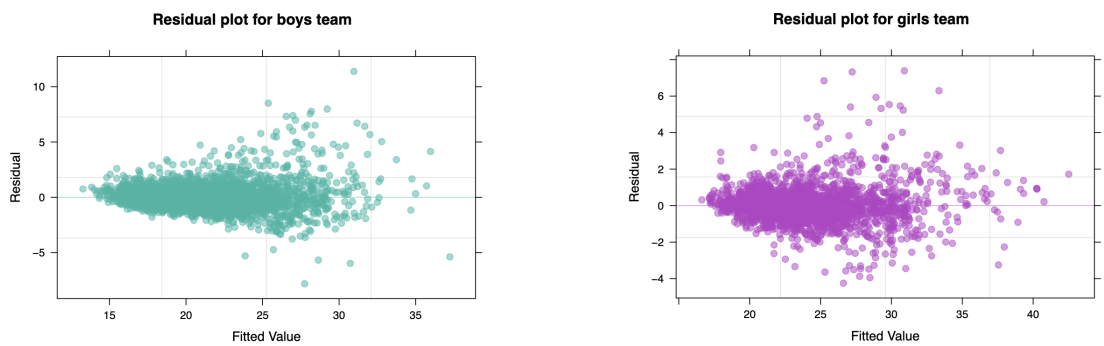

Figure 4: Residual versus fitted plots for both teams. Slower times tend to have larger residuals.

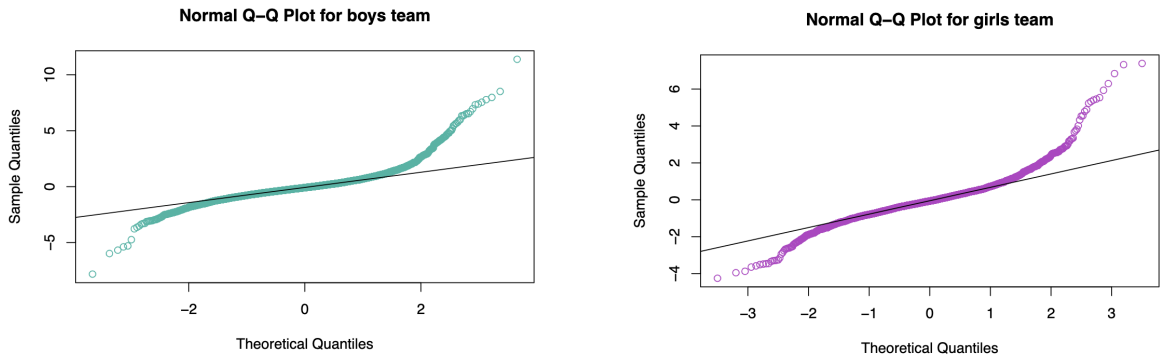

Figure 5: Q-Q plots for both teams. The center of the distribution is alright, but the tails diverge heavily.

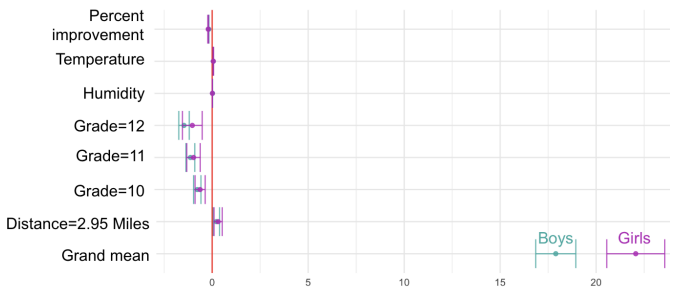

Figure 6: Fixed effects for non-year variables (95% confidence).

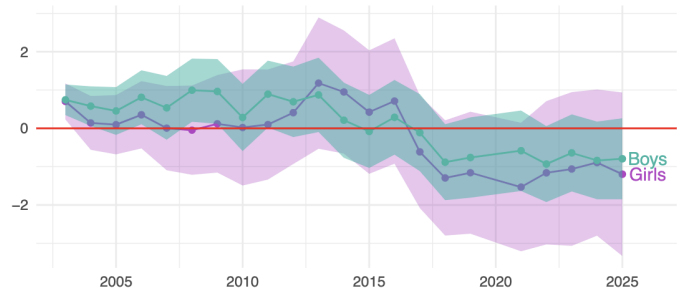

Figure 7: Fixed effects for year, with 2002 as baseline (95% confidence).

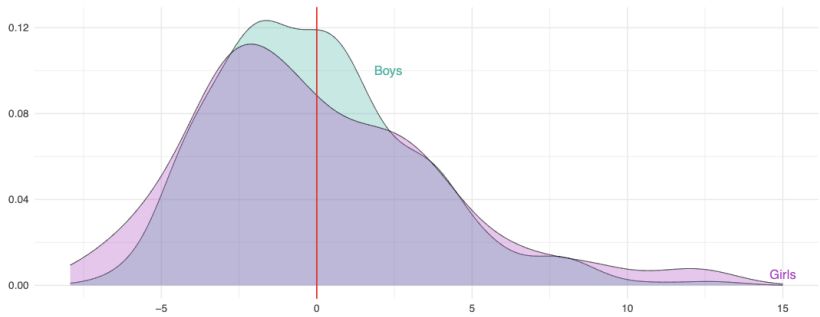

Figure 8: Distribution of the random intercepts for athlete.

4 Discussion

The fixed effects for percent improvement, temperature, and humidity, while significant at a p-value of 0.05, are close to 0 (Figure 6). Increases in temperature and humidity do seem to correspond to slower times, although again this effect is small. This does make sense, as it is harder to run when the weather is more extreme. However, the fixed effects for grade, distance, and the grand mean are all significantly not 0 at the 0.05 level (6.3). Increasing grade is associated with faster times. Aside from the grand mean of the boys team being faster than the girls team, and grade having a slightly larger effect on boys' times compared to girls', the predictors actually do not differ that much by gender.

None of the fixed effects by year appear to be significant at the 0.05 level. However, observing the confidence interval plots in Figure 6 and 7, the fixed effects for the years on the girls team has a wider 95% confidence interval than the boys, with the width also increasing the farther one gets from the baseline of 2002. This may be because the boys team has more results and less variance in their times generally (Figure 12). Both teams exhibit similar trends, with there appearing to be a shift of the teams

tending to be faster rather than slower compared to 2002, starting around 2017.
Additionally, the random intercepts for athlete have a good amount of variance, with the girls team also appearing to have a larger variance than the boys (Figure 8).
As seen in the residual plots and Q-Q plots in 3, the model residuals for both teams are heteroskedastic, with the variance tending to increase as the predicted time in minutes increases, and the data also exhibits non-normality with heavy tails, especially on the right side. This can be explained by the fact that the team does not typically make cuts based on performance, and athletes on the reserve team are not under pressure to score points. To address these violations, other modeling approaches were attempted, including taking the log of the times and using a generalized linear mixed effects model with a Gamma distribution on the times. They were ultimately scrapped as they did not fully resolve these issues and were more difficult to interpret.

4.1 Limitations

The data collected is purely observational, thus making it difficult to determine any causal relationships. In addition, the data has a few other restrictions. The weather data is not exact to the time or location of a race, instead being averaged across an approximate window of time when races would take place and being drawn from a nearby weather station. The data only includes results and athletes at two distances; even though these can be argued to make up the majority, they still leave out some results and athletes. Finally, this data is specific to this school and may not be generalizable to other schools.

5 Conclusion

In conclusion, there is no significant difference in results at each course between years, while an increase in grade decreases times and increases in temperature and humidity have a small effect on increasing times. The failure of the model assumptions is most likely due to the fact that the data is very non-normal (Figure 11). In terms of the variables' predictive power, statistics like age, height and weight on each athlete, which may be more strongly associated with the times, are not available on athletic.net.

6 Appendix

6.1 AI Disclosure Statement

AI was used for portions of the coding for this project, mostly to edit graphs for display, and to provide feedback on the modeling direction. Specifically, ChatGPT version 5.2 was used. Full transcripts of the interactions can be found at these links: `https://chatgpt.com/share/6994fcfd-0ec8-8005-8d1d-01633895339b` `https://chatgpt.com/share/69a3b0e4-3ad4-8005-9983-d06110352abe`

6.2 EDA

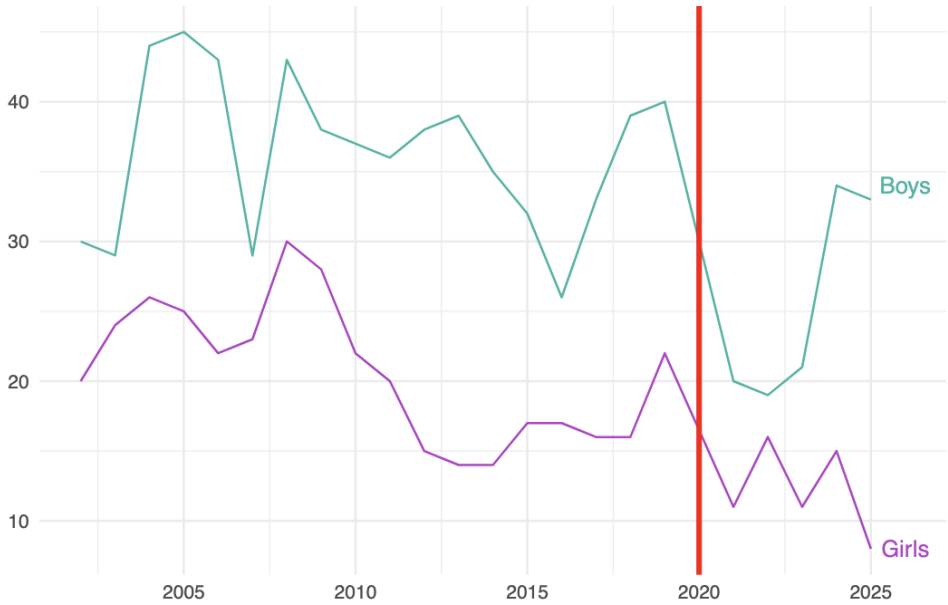

Figure 9: The boys team always has had more members than the girls. Interestingly, both teams saw a significant drop in size after COVID-19, but the boys team was able to recover by 2024, while the girls team did not.

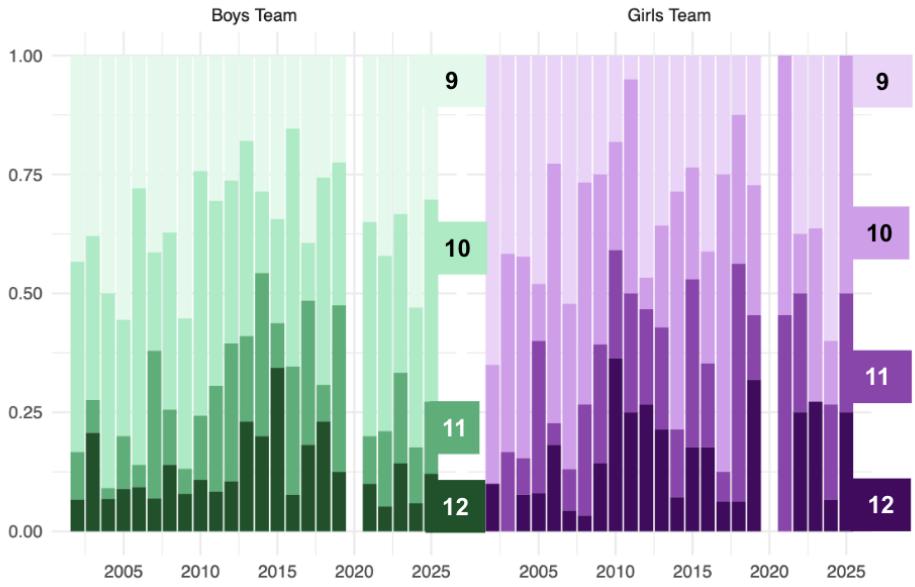

Figure 10: Grade distributions for both teams by year. Underclassmen (grades 9 and 10) make up the majority of both teams. One possible reason for the recovery of the boys team size post-COVID is the relative constant proportion of freshmen, as the girls team has hit zero freshmen twice in the last five years.

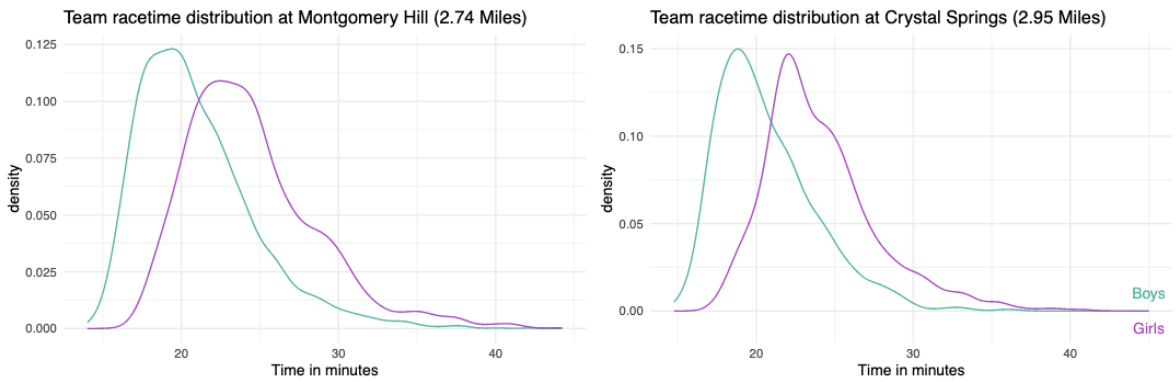

Figure 11: Clear differences in the distribution of times between the boys and girls teams. Also showing the data being significantly skewed right and non-normal.

	2.74 Miles		2.95 Miles	
Characteristic	Boys	Girls	Boys	Girls
Number of results	2821	1503	1097	661
Grade: 9	1000 (35.45%)	472 (31.40%)	371 (33.82%)	210 (31.77%)
Grade: 10	995 (35.27%)	531 (35.33%)	348 (31.72%)	218 (32.98%)
Grade: 11	462 (16.38%)	306 (20.36%)	222 (20.24%)	134 (20.27%)
Grade: 12	364 (12.90%)	194 (12.91%)	156 (14.22%)	99 (14.98%)
Time in minutes (Median (IQR))	20.27 (18.23–22.85)	23.72 (21.46–26.60)	20.02 (18.34–22.41)	23.36 (21.71–25.80)
% results that were PR's	16.31%	16.03%	32.91%	27.84%
Grade: 9	16.00%	19.70%	35.85%	29.52%
Grade: 10	18.09%	16.57%	37.64%	33.94%
Grade: 11	12.12%	12.42%	22.52%	23.13%
Grade: 12	17.58%	11.34%	30.13%	17.17%
Temperature by course (F) (Median (IQR))	75.97 (70.70–81.58)		63.74 (60.80–66.76)	
Humidity by course (%) (Median (IQR))	42.00 (32.92–50.00)		61.40 (53.15–73.45)	

Figure 12: Summary statistics of some of the predictors, also including PR rates per grade. The temperature and humidity both tend to be higher at the 2.74 mile course than the 2.95 mile course. It appears comparatively easier to PR at 2.95 miles, but this is most likely due to the teams racing less often at this distance.

6.3 Model Coefficients

Table 1: Model coefficient estimates for boys team

	Estimate	Std. Error	t value
(Intercept)	17.8966774	0.4834470	37.0188991
grade10	-0.7758720	0.0662802	-11.7059353
grade11	-1.1341956	0.0885756	-12.8048355
grade12	-1.4662231	0.1081785	-13.5537341
time_pct_imp	-0.1796493	0.0032434	-55.3896839
temp	0.0561969	0.0037737	14.8918058
rhum	0.0167587	0.0018774	8.9266384
race_year2003	0.7437296	0.2552685	2.9135193
race_year2004	0.5792477	0.2804021	2.0657753
race_year2005	0.4525910	0.2927822	1.5458282
race_year2006	0.8099098	0.3162995	2.5605786
race_year2007	0.5344353	0.3369921	1.5858988
race_year2008	0.9945192	0.3477490	2.8598764
race_year2009	0.9621034	0.3619369	2.6582074
race_year2010	0.2832790	0.3740462	0.7573368
race_year2011	0.8922949	0.3876510	2.3017993
race_year2012	0.6932464	0.3941481	1.7588475
race_year2013	0.8726597	0.4114885	2.1207391
race_year2014	0.2125407	0.4216866	0.5040253
race_year2015	-0.0813198	0.4343280	-0.1872314
race_year2016	0.2875143	0.4491080	0.6401896
race_year2017	-0.1118600	0.4514170	-0.2477974
race_year2018	-0.8838886	0.4637726	-1.9058665
race_year2019	-0.7618356	0.4789364	-1.5906820
race_year2021	-0.5838588	0.5022632	-1.1624559
race_year2022	-0.9322168	0.5208280	-1.7898745
race_year2023	-0.6412978	0.5294761	-1.2111931
race_year2024	-0.8377068	0.5398240	-1.5518146
race_year2025	-0.7943165	0.5644474	-1.4072464
distance2.95 Miles	0.2150691	0.0586786	3.6652026

Table 2: Model coefficient estimates for girls team

	Estimate	Std. Error	t value
(Intercept)	22.0617118	0.6428646	34.3178215
grade10	-0.6291531	0.0919980	-6.8387673
grade11	-0.9714870	0.1233899	-7.8733126
grade12	-1.0340557	0.1613399	-6.4091739
time_pct_imp	-0.2084396	0.0043253	-48.1913192
temp	0.0591997	0.0048783	12.1351875
rhum	0.0114593	0.0022549	5.0820103
race_year2003	0.7015115	0.2467928	2.8425123
race_year2004	0.1394345	0.2997420	0.4651819
race_year2005	0.0953854	0.3278417	0.2909495
race_year2006	0.3529340	0.3739483	0.9438041
race_year2007	0.0046734	0.4040104	0.0115676
race_year2008	-0.0484037	0.4316992	-0.1121236
race_year2009	0.1179807	0.4588659	0.2571137
race_year2010	0.0214907	0.4905527	0.0438092
race_year2011	0.0980271	0.5273585	0.1858833
race_year2012	0.4082124	0.5576120	0.7320725
race_year2013	1.1797861	0.5836695	2.0213257
race_year2014	0.9485122	0.6185043	1.5335581
race_year2015	0.4222247	0.6426319	0.6570242
race_year2016	0.7113830	0.6620452	1.0745233
race_year2017	-0.6141224	0.6783098	-0.9053716
race_year2018	-1.2946039	0.7082972	-1.8277692
race_year2019	-1.1586830	0.7286874	-1.5900960
race_year2021	-1.5337304	0.8009570	-1.9148722
race_year2022	-1.1603639	0.8175434	-1.4193300
race_year2023	-1.0613525	0.8625607	-1.2304671
race_year2024	-0.8935829	0.8998720	-0.9930110
race_year2025	-1.1967715	0.9520089	-1.2571012
distance2.95 Miles	0.3107580	0.0759030	4.0941460

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